Chemical Engineering Project Portfolio

Emmanuel Galindo-Montellano
UC San Diego — Pharmacological Chemistry
Expected Graduation: 2026

Core Engineering Areas

- Reaction Engineering and Kinetics
- Separation Processes and Phase Equilibria
- Bioprocess Engineering
- Environmental Chemical Analysis
- Process Data Analysis and Modeling

Technical Skills

Protein Purification • SDS-PAGE • Enzyme Kinetics • Gas Chromatography • UV-Vis Spectroscopy • Atomic Absorption Spectroscopy • Distillation • Data Analysis • MATLAB

Introduction

This portfolio presents laboratory and computational projects demonstrating core chemical engineering principles including reaction kinetics, separation processes, bioprocess engineering, and reactor modeling. These projects integrate experimental analysis with quantitative modeling and illustrate the application of engineering concepts to biochemical and environmental systems.

Included Projects

1. Enzyme Kinetics Modeling of Lactate Dehydrogenase
2. Recombinant Protein Expression and Purification
3. Fractional Distillation and Gas Chromatographic Analysis of Hydrocarbon Mixtures
4. Quantitative Environmental Chemical Analysis of Marine Water Systems

Project 1

Reaction Kinetics Analysis of an Enzyme-Catalyzed System

Objective

Characterize the catalytic kinetics of lactate dehydrogenase and evaluate the effect of inhibitor concentration on enzyme activity.

Experimental Method

Reaction rates were measured by monitoring the oxidation of NADH at 340 nm using UV-Vis spectrophotometry. Reaction velocity was determined across varying pyruvate concentrations with and without inhibitors.

Michaelis–Menten Analysis

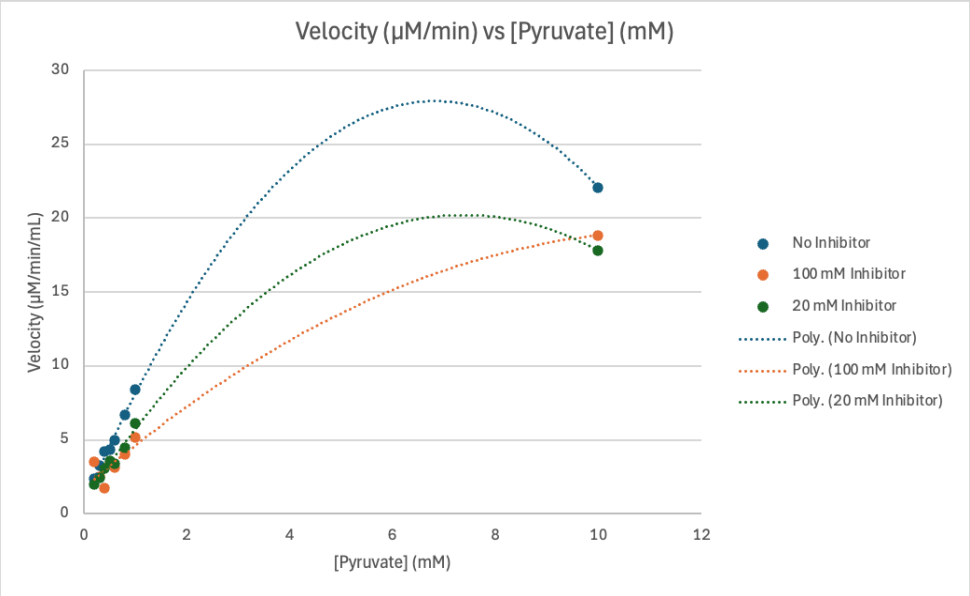


Figure 1. Reaction velocity of lactate dehydrogenase as a function of pyruvate concentration measured using UV-Vis spectrophotometry at 340 nm. Increasing substrate concentration results in higher reaction velocity until enzyme active sites become saturated, demonstrating characteristic Michaelis–Menten kinetic behavior. The presence of inhibitor reduces reaction velocity across substrate concentrations, indicating suppression of catalytic activity.

Observations

- Reaction velocity increases with substrate concentration
- Saturation behavior indicates enzyme kinetics

- Inhibitor presence reduces reaction velocity

Lineweaver–Burk Analysis

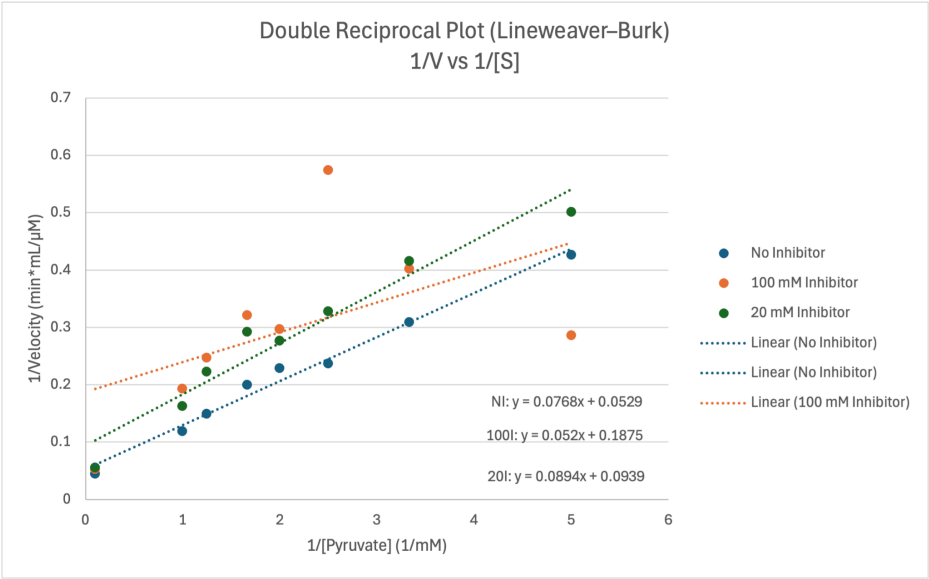


Figure 2. Lineweaver–Burk double reciprocal plot (1/V vs 1/[S]) used to estimate kinetic parameters for lactate dehydrogenase under different inhibitor concentrations. Linear regression of the transformed data provides estimates of the Michaelis constant (Km) and maximum reaction velocity (Vmax). Changes in slope and intercept reflect the influence of inhibitor concentration on enzyme kinetics.

Regression equations:

No inhibitor
$y = 0.0768x + 0.0529$

100 mM inhibitor
$y = 0.052x + 0.1875$

20 mM inhibitor
$y = 0.0894x + 0.0939$

Estimated parameters:

$V_{max} \approx 18.9$ µM/min
$K_m \approx 1.45$ mM

Engineering Significance

Reaction kinetics govern the design of chemical and biochemical reactors. Determining kinetic parameters such as K_m and V_{max} allows engineers to model reaction rates and optimize reactor operating conditions. Enzyme kinetics are widely applied in bioreactors used in pharmaceutical production, fermentation processes, and industrial biocatalysis.

Project 2

Downstream Processing and Chromatographic Separation of Recombinant Protein

Objective

Develop a purification workflow for recombinant LuxG flavin reductase expressed in *E. coli*.

Bioprocess Downstream Separation Workflow

E. coli Culture
↓
Protein Induction (IPTG)
↓
Cell Lysis
↓
Ammonium Sulfate
↓
Precipitation
↓
Dialysis
↓
Ni-Affinity Chromatography
↓
Purified Enzyme

Protein Verification

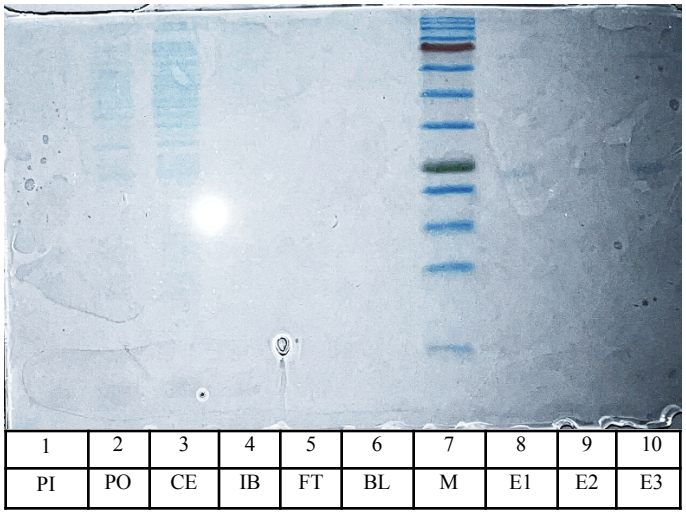


Figure 3. SDS-PAGE analysis of recombinant LuxG purification from E. coli. Lane assignments: (1) pre-induction control, (2) post-induction sample after IPTG expression, (3) crude cell extract following sonication, (4) inclusion body fraction, (5) column flow-through, (6) buffer blank, (7) molecular weight ladder, (8) elution fraction E1 (50 µg), (9) elution fraction E2 (50 µg), and (10) elution fraction E3 (75 µg). A prominent band observed in the elution fractions corresponds to LuxG with an estimated molecular weight of approximately 24–25 kDa based on interpolation from the semi-log calibration curve. The enrichment of this band in the elution fractions indicates successful purification of the His-tagged LuxG protein using Ni-affinity chromatography.

Interpretation

- Multiple bands in crude extract
- Enrichment of target protein band following purification
- Molecular weight consistent with LuxG protein

Engineering Significance

Downstream processing is a critical stage in biopharmaceutical manufacturing. Separation techniques such as precipitation, dialysis, and affinity chromatography are used to isolate and purify target biomolecules from complex mixtures. Understanding these processes is essential for designing large-scale purification systems in industrial biotechnology.

Project 3

Vapor–Liquid Separation of Hydrocarbon Mixtures Using Fractional Distillation

Objective

Separate a binary hydrocarbon mixture using fractional distillation and evaluate composition using gas chromatography.

Separation Principle

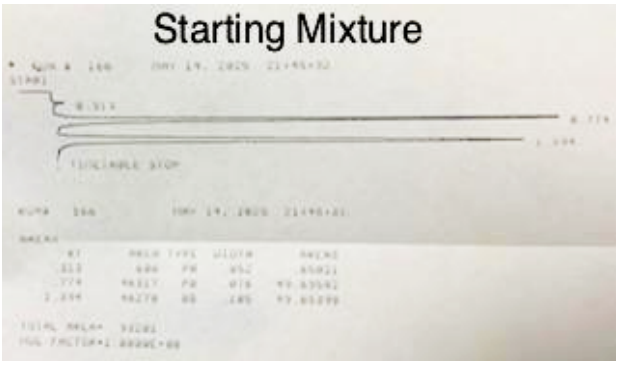
Fractional distillation separates compounds based on differences in boiling point and vapor pressure.

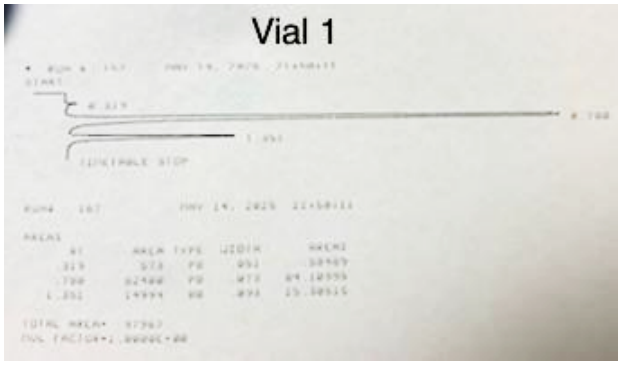
Hexane boiling point: ~69 °C
Heptane boiling point: ~98 °C

Mass Transfer and Phase Equilibrium Considerations

Fractional distillation relies on vapor–liquid equilibrium and repeated vaporization–condensation cycles to achieve separation. The process efficiency depends on boiling point differences, column design, and the number of theoretical plates. These principles are fundamental in industrial distillation columns used in petroleum refining and chemical manufacturing.

Gas Chromatography Analysis





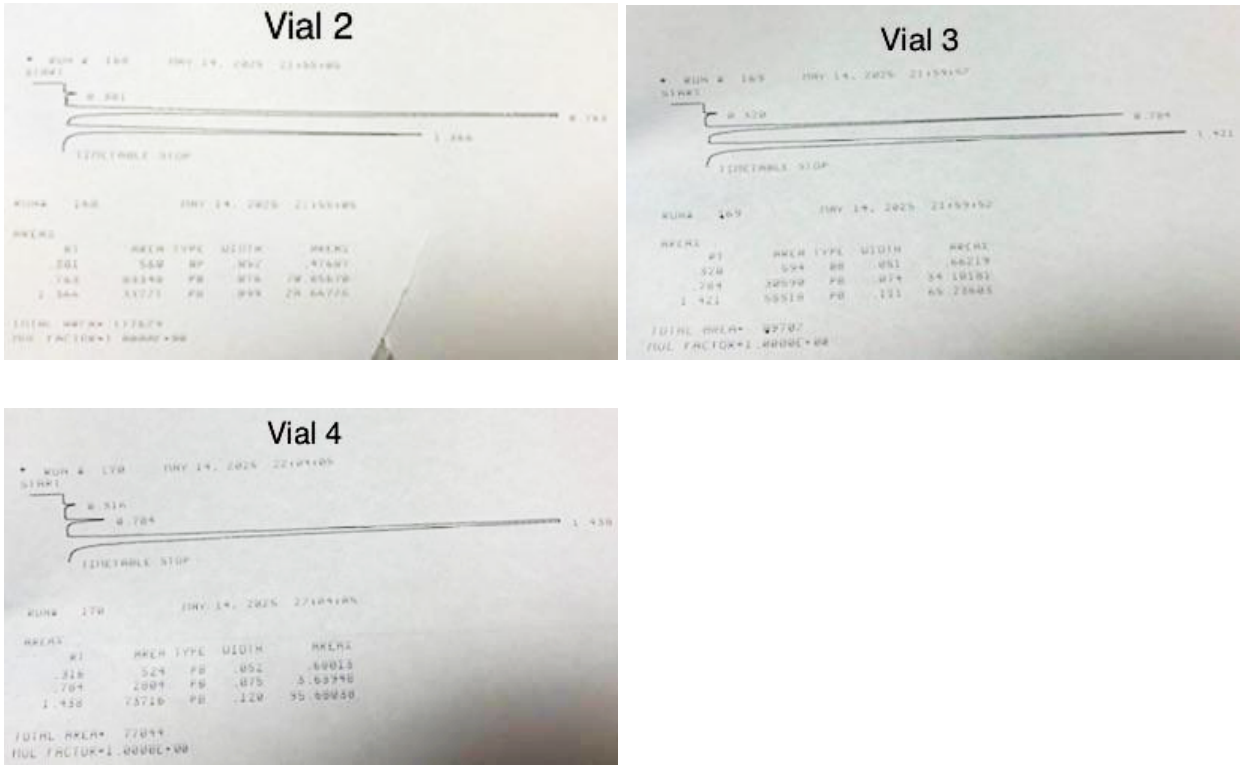


Figure 4. Gas chromatography chromatograms of fractions collected during fractional distillation of a hexane–heptane mixture. Early fractions exhibit dominant peaks corresponding to hexane due to its lower boiling point (~69 °C), while later fractions show increased heptane composition (~98 °C). The chromatograms demonstrate progressive separation of the binary mixture as distillation proceeds.

Observations

- Early fractions enriched in hexane
- Later fractions enriched in heptane
- Two peaks correspond to individual hydrocarbon components

Engineering Significance

Distillation is the most widely used industrial separation process in chemical manufacturing, petroleum refining, and solvent purification.

Project 4

Quantitative Environmental Chemical Analysis of Marine Water Systems

Objective

Quantitatively determine the concentration of multiple chemical species in seawater samples using analytical chemistry instrumentation and calibration-based measurement techniques.

Experimental Method

Multiple analytical techniques were applied to characterize seawater composition collected from the Scripps Pier in La Jolla, California.

UV–Vis Spectrophotometry

Reactive orthophosphate concentration was determined using the molybdenum blue reaction. The resulting phosphomolybdate complex was analyzed spectrophotometrically using a Genesys 30 spectrophotometer to determine absorbance at λmax. External calibration curves were constructed from standard phosphate solutions to quantify the unknown sample concentration.

Flame Atomic Absorption Spectroscopy

Calcium concentration was measured using flame atomic absorption spectroscopy. Standard calcium solutions were prepared to generate an external calibration curve relating absorbance to Ca^{2+} concentration, allowing determination of the calcium concentration in seawater samples.

Winkler Titration

Dissolved oxygen concentration was determined using the Winkler iodometric titration method. Oxygen in the seawater sample was chemically fixed using manganese reagents and quantified through titration with sodium thiosulfate.

Acid–Base Titration and Gran Plot Analysis

Seawater alkalinity was determined through titration with standardized hydrochloric acid. A Gran plot was used to determine the equivalence point and calculate the total alkalinity of the seawater sample.

Representative Results

Analyte	**Method**	**Result**
Phosphate	UV-Vis Spectroscopy	0.969 ± 0.018 μmol/kg

Calcium	Atomic Absorption Spectroscopy	435 ± 41 ppm
Dissolved Oxygen	Winkler Titration	0.0002401 M ± 1.2E-05 M
Alkalinity	Gran Plot Analysis	0.0029 ± 0.0001 mol/kg

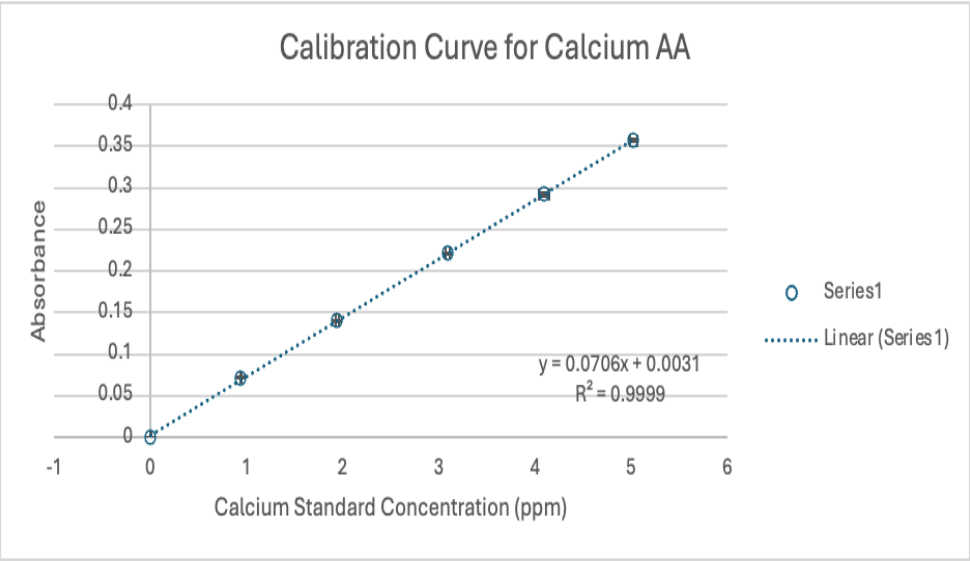


Figure 5. Calibration curve for calcium concentration determined using flame atomic absorption spectroscopy. Standard calcium solutions were used to generate a linear relationship between absorbance and concentration, allowing quantitative determination of calcium concentration in seawater samples through external calibration.

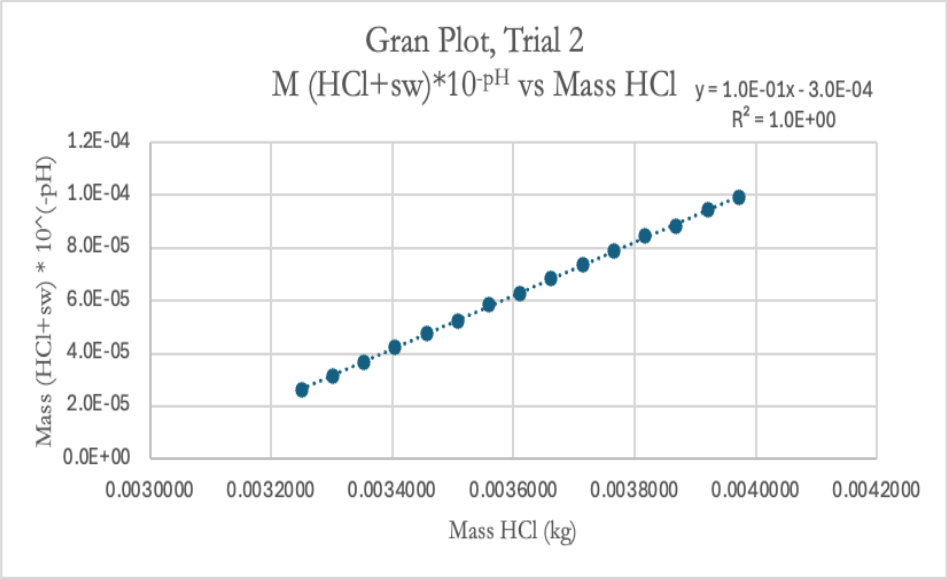


Figure 6. Gran plot used to determine the equivalence point during acid titration of seawater for total alkalinity measurement. The linear extrapolation of the transformed titration data enables precise identification of the equivalence point and calculation of alkalinity concentration in the sample.

Engineering Significance

Water quality monitoring requires quantitative analytical techniques to evaluate chemical species in natural water systems. These measurements are essential in environmental engineering for monitoring pollution, evaluating treatment systems, and designing water purification processes.

Project 5

Dynamic Modeling and Simulation of Hydrogen Peroxide Decomposition in a Continuous Stirred Tank Reactor Using MATLAB

Objective

Develop a computational model to simulate the decomposition of hydrogen peroxide in a continuous stirred tank reactor and analyze how operating conditions influence reactant conversion and reactor performance.

Reaction System

Hydrogen peroxide decomposes into water and oxygen according to the reaction:

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

Rate law:

$$-r_{H2O2} = k[H_2O_2]$$

- k = reaction rate constant
- $[H_2O_2]$ = Hydrogen peroxide concentration

Reactor Model

The reactor considered in this project is a continuous stirred tank reactor operating under steady flow conditions.

A transient mass balance for hydrogen peroxide in the reactor is:

$$\frac{dC_A}{dt} = \frac{F}{V}(C_{A0} - C_A) - kC_A$$

- C_A = hydrogen peroxide concentration inside reactor
- C_{A0} = inlet concentration
- F = volumetric flow rate
- V = reactor volume
- k = reaction rate constant

This equation accounts for:

• convective transport from the inlet stream
• reaction consumption of hydrogen peroxide inside the reactor

Simulation Approach

The differential mass balance equation was solved numerically using MATLAB to simulate the dynamic concentration profile within the reactor.

Simulations were used to evaluate:

- concentration of hydrogen peroxide over time
- reactant conversion behavior
- steady-state reactor performance

Operating parameters used in the model included assumed feed concentration, reactor volume, and flow rate representative of laboratory-scale reactor conditions.

Results

Concentration Dynamics

Simulation results show that hydrogen peroxide concentration decreases as the reaction proceeds within the reactor until a steady state is reached.

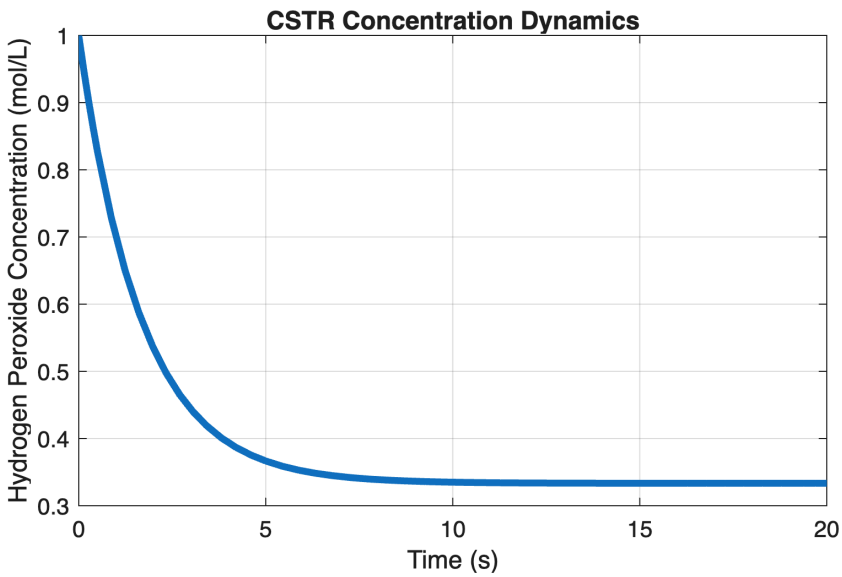


Figure 7. Transient hydrogen peroxide concentration in a continuous stirred tank reactor obtained through numerical solution of the reactor mass balance using MATLAB. The concentration decreases over time as the decomposition reaction proceeds until steady-state conditions are reached where inflow, outflow, and reaction consumption are balanced. Simulation parameters: reaction rate constant **k = 0.4 s^{-1}**, reactor volume **V = 5 L**, volumetric flow rate **F = 0.1 L s^{-1}** , and inlet hydrogen peroxide concentration **C_{A0} = 1.0 mol L^{-1}.**

Reactant Conversion

Conversion of hydrogen peroxide is defined as:

$$X = \frac{C_{A0} - C_A}{C_{A0}}$$

Simulation results show that conversion increases as residence time increases, reflecting greater exposure of reactant to the reaction environment.

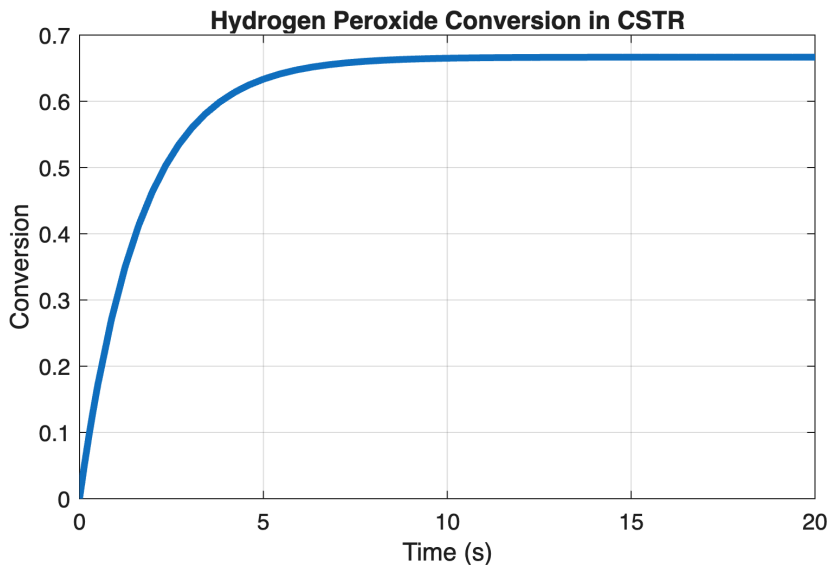


Figure 8. Conversion of hydrogen peroxide as a function of time for a continuous stirred tank reactor simulated using MATLAB. Conversion increases as the reaction progresses and approaches a steady-state value determined by the reactor residence time and kinetic parameters governing the decomposition reaction.

Conversion vs Residence Time

Chemical engineers immediately recognize this because reactor performance is usually analyzed in terms of residence time.

$$\tau = \frac{V}{F}$$

- V = reactor volume
- F = volumetric flow rate

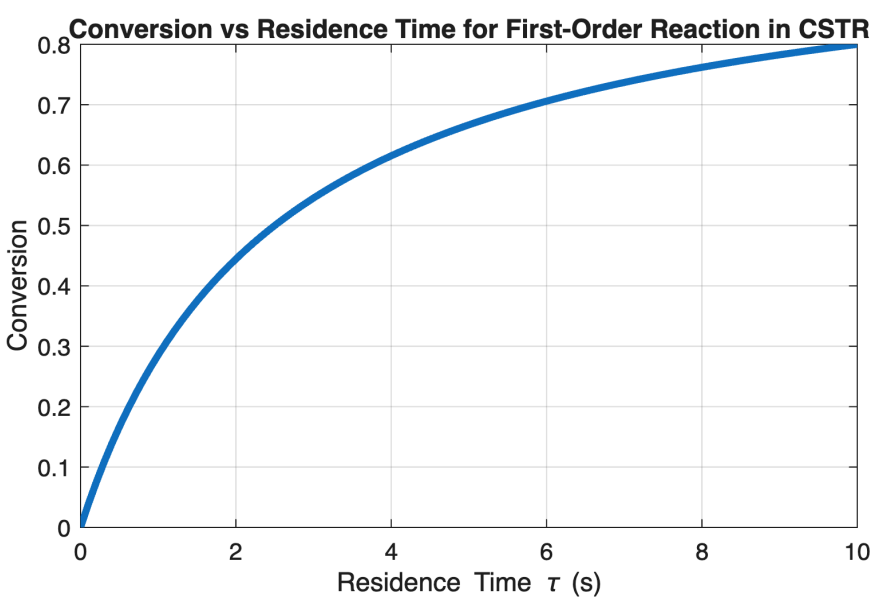


Figure 9. Conversion of hydrogen peroxide as a function of reactor residence time for a first-order reaction in a continuous stirred tank reactor. Increasing residence time allows reactants to remain in the reactor longer, resulting in greater conversion. The curve illustrates the relationship between reactor residence time and performance for first-order reaction kinetics.

Engineering Significance

Chemical reactors are fundamental components of industrial chemical production. Modeling reactor behavior allows engineers to predict reactant conversion, optimize operating conditions, and design reactors that maximize product yield.

Hydrogen peroxide decomposition provides a useful model system for studying reactor dynamics because its kinetics are well understood and the reaction is relevant to applications including wastewater treatment, chemical synthesis, and propulsion systems.

Computational reactor modeling is widely used in chemical engineering to analyze reaction systems before pilot-scale or industrial implementation.

Technical Summary

Core Engineering Skills:

Reaction Engineering

- Reaction kinetics modeling
- Enzyme catalysis analysis
- CSTR dynamic simulation

Separation Processes

- Fractional distillation
- Chromatographic purification
- Gas chromatography
- Analytical Techniques
- UV-Vis spectroscopy

Atomic absorption spectroscopy

- SDS-PAGE protein analysis
- Computational Tools
- MATLAB numerical simulation
- Data analysis and regression
- Calibration curve development

These skills are applicable to fields including:

Bioprocess engineering
Pharmaceutical development
Analytical chemistry
Chemical process design